CIVIL MAINT. Department
SIMHADRI

Ref:SMTPP/Civil Maint/2022-23/NS/326708

Date: 02/08/2022(15:57:56)

Subject:LMI/OPS/CIVIL/002-Power Station Drainage. Rev.02, Approval for implementation-reg.

LMI/OPS/CIVIL/002, Rev.2 has been prepared based on revised OGN ref. COS-ISO-00-OGN/ENG/002 and is attached herewith.

The same may please be approved by competent authority for implementation.

Submitted by:

जयंती गोपी किरण - उप महाप्रबंधक

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May please be approved

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Revised portions if any, may please be mentioned/highlighted before going for further approval. Also since the responsibiltiy centres are clearly defined/tabulated on page no 14, Clause no. 7, of the LMI, please route (also please modify the File routing Path) the LMI through the concerned Department / Sections before final approval.

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Pl comply

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Clause no. 4.3.5 & 4.3.6 has been modified as per the revised OGN.

File routed through concerned Department / Sections before final approval by competent authority.

Submitted for approval please.

Submitted by:

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Submitted for approval pls

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Review and comment on clause 4.3.5

Submitted by:

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On:16/08/2022(17:27:08) **Ref:**SMTPP/Civil Maint/2022-23/NS/326708

In NTPC-SIMHADRI for GT and other power transformers cooling , water is not used in cooling circuit , hence no need of separate arrangement for ARCW water (for GT) as stated in in LMI .

Submitted by:

सतीश कुमार सिंह - उप महाप्रबंधक

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May please be accorded approval.

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On:23/08/2022(18:25:09) **Ref:**SMTPP/Civil Maint/2022-23/NS/326708

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

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NTPC **SIMHADRI**

<u>**LOCATION MANAGEMENT INSTRUCTION**</u>

LMI/OPS/CIVIL/002
Rev. No: 02; Aug' 2022

LMI FOR POWER SYSTEM DRAINAGE

Approved for
Implementation ....
HOP, SIMHADRI

Date :....................................

LMI APPROVAL DOCUMENTS

1. TITLE OF LMI	LMI FOR POWER SYSTEM DRAINAGE
2. a) DOCUMENT CODE b) REVISION/ ISSUE NO.&DATE	LMI/OPS/CIVIL/002 Rev. No.: 02; Aug'2022
3. ORIGINAL/ REF.DOCUMENT	COS-ISO-00-OGN/ENG/002 Issue No.: 1, Date: MAY'2022
4. PREPARED BY	Name: Shri. PVSN RAJU Mgr. (O&M-Civil)
5. CHECKED BY	Name: Shri. J. Gopi Kiran DGM (O&M-Civil)
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7. APPROVED BY	Name: Shri. Girish Chandra Choukse HOP (Simhadri)
8. DATE OF ISSUE	
9. INDEX LIST	Enclosed

POWER STATION DRAINAGE

CONTENTS

POWER STATION DRAINAGE

1.0 INTRODUCTION

This document gives guidance on specific aspects of drainage required on Simhadri power station sites.

This document does not include transformer compound drainage or coal store drainage which are dealt with in other documents.

2.0 SUPERSEDED DOCUMENT:

OGN-ENG-002 ISSUE 1, OCT. 1994, REVIEWED ON APRIL 2020

3.0 DRAINAGE OF SURFACE WATER AND FOUL WATER

3.1 GENERAL REQUIREMENTS

Drainage systems should be self-cleansing. Systems collecting surface water run-off should be designed for a rainfall of 1 in 50 years intensity with pipes running un-surcharged at self cleaning/non scouring velocities. Changes of gradient, the number of bends, traps, syphons etc, should be kept to a minimum. Access must be provided so that it is possible to rod the whole system.

Pipe drains are considered for the storm water drainage. However, for power stations, where large surface areas are to be drained, the open drains have following distinct advantages.

a) Economy

b) Ease of inspection & maintenance

c) Free Board margin gets utilized in excess floods.

d) Require less earth work in plant levelling, where plants are on filled up areas.

Accordingly open drains are being considered in the design by NTPC.

Laying drains under buildings should be avoided where possible. Drain runs should be located in road verges rather than under roads.

Where there is the possibility of contamination of surface water by oil, the water draining from that area must be passed through an oil/water separator before discharge.

3.2 MANHOLES AND CATCH PITS

Manholes should be provided at all changes of direction with gradient and distance between manholes as per the relevant Indian standards.

Catch pits should be used only for surface water and plant water drainage when a high proportion of solids is expected, as beneath precipitator hoppers. The flow cross sectional area should be large enough to slow down the flow velocity by a factor of at least ten to allow solids to be deposited and to provide adequate capacity for the solids. As far as possible large catch pits should be so located that access to them from a road or paved area is easy, to permit cleaning.

3.3 GULLEY POTS

Road gulley pots with a nominal diameter of 450mm and depth of 900mm are large enough to preclude the necessity for frequent cleaning out for all but extremely dirty conditions. Gully pots should be trapped and fitted with a rodding eye. Road gullies should be located so that the maximum width of flow under design storm condition is 750mm.

3.4 ROOF DRAINAGE

Where it is necessary to route rainwater pipes inside station buildings, special consideration should be given to the design of any pipes which pass through an area containing electrical equipment and the provision of a watertight housing or sleeve would be appropriate.

3.5 FOUL DRAINAGE CAPACITY

The foul water flow for ablutions/toilets shall be taken as per the relevant Indian standard codes.

4.0 DRAINAGE INSIDE PLANT BUILDINGS

4.1 DRAINAGE FROM ENGINEERING PLANT

The quantity of waste water to be discharged may be assessed from plant schematics and plant contractor's estimates. Water used cool bearings, etc is normally recirculated but may be discharged to drains when quantities are relatively small or cannot be recovered.

Maximum discharge will occur during normal operation and the worst condition should be considered in each section of the drainage system for (a) normal running including starting conditions (b) one unit being drained and the remainder running.

In areas where water spray fire protection systems are to be installed, eg. cable races, drainage should be provided within reason. It may be necessary to tank the floor if there is a plant area below which is sensitive to water percolation.

4.2 BOILER HOUSE DRAINAGE

4.2.1 ASH SLUICE AREA

Drain channels covered by open grids should be provided. Where the disposal of smaller quantities of water in the area cannot be easily accommodated by drains it may be possible to discharge these into the ash sluice trench.

4.2.2 PULVERISED FUEL MILLS FLOOR AREA

Drainage is required in a pulverized fuel mill area for washing down which, in practice, takes place once a day.

Washing down will also be required in the event of a pulverized fuel leak and this can result in the water containing a considerable proportion of solids.

4.2.3 BOILER WASHING

The effluent from the gas-side washing down of boiler components may be acidic, contain heavy mental contaminants and, in the case of pulverized fuel firing, contain fine ash which may block drains.

Initial washings will be discharged to a sump for removal by tanker. Late washings may be clean enough to be discharged into the station drainage system.

The economizer and air heaters may be washed down independently and initial washings will be discharged to a holding sump. Water jets are installed in the air heaters for quenching possible fires.

Where there is the possibility of spillage during the washing process, the boiler house floor should be provided with drainage points for washing down afterwards.

4.2.4 CLEANING OF BOILERS AND FEED SYSTEMS BEFORE COMMISSIONING

During construction at least one hydraulic test will be carried out on the boiler pressure parts. After testing the water must be drained away in reasonable time, say not more than 6 hours. The same procedure must be used after a pressure part repair during the life of the boiler.

During cleaning, boilers are filled with acid or alkali and flushed through three or four times with water. The following methods have been used for the disposal of effluent:

i) The chemical effluent is diluted with water and discharged to the plant drainage system.

ii) It is left to stand for a period in a borrow pit formed during the construction of the station or discharged to an effluent lagoon where this exists.

4.2.5 BOILER BLOWDOWN

Because of the general developments in boilers the following points should be carefully considered during the design of drainage:

There are a number of blowdown points and although water is usually recovered for further use some may be discharge through a blowdown vessel to the drains. It may be discharged with equal quantities of cooling water from the auxiliary circulation water cooling system. When this is not done the water will be nearly boiling. Glazed earthenware pipes can crack at this temperature and cast iron pipes should be used of this type of discharge. The drain should be covered but vapour release must be allowed for.

4.3 TURBINE HOUSE DRAINAGE

4.3.1 CONDENSATE EXTRACTION PUMP

Provision should be made for collecting any leakage of cooling water which may occur through glands, and for a possible small discharge of sealing water.

4.3.2 AIR EXTRACTION PLANT

In normal operation the quantity of cooling water circulating around the air pumps is in the region of 1.5 litres/s/KW. The quantity of water to be disposed off and discharged into the drainage system will, however, vary according to design. Where the quick-start air pump is installed and overflow of water occurs for a short period while the set is starting up.

4.3.3 BLEED STEAM EVAPORATOR PLANT

There is continuous blowdown from evaporation, which may contain some solids. The drains should be enclosed to prevent vapour from the discharge causing damage to adjacent electrical gear.

4.3.4 CONDENSATE (DISTILLATE) DRAIN

On start-up the condensate may be drained to waste until the quality is satisfactory. On commissioning, the flow may last for several days, and the amount of drainage would be less on subsequent start-ups. Although some of the condensate may be large enough to affect the drain size which should be adequate to receive the anticipated flow.

4.3.5 AUXILIARY CIRCULATING WATER COOLING SYSTEM

The auxiliary circulating water cooling system, which can be emptied to the drains often consists of one branch from the main circulating water system which passes though all the coolers associated with one unit. A strainer is fitted near to the beginning of the auxiliary circulation water cooling system. The strainer can also be emptied, and flushing water may be passed from the auxiliary circulating water system to the drains. The following items may occur in one circuit:

- i) Main turbine oil coolers
- ii) Main boiler feed pumps
- iii) Seal oil coolers
- iv) Exciter air coolers
- v) Distilled water coolers

vi) Air extraction pumps
vii) Generator transformer oil coolers
viii) Turbine block cooler- where applicable.

Drain gullies are required for the occasional emptying of cooler, but the flows and quantities are small. However, in some instances, water for generator transformer cooling taken form the ACW cooling system cannot be recirculated and has to be discharged to the ZLD sump.

4.3.6 COOLING WATER IN CONDENSER

Several types of condenser have been developed. The operational requirement for the draining of the condenser should be verified together with the number of condensers or sections of condensers to be drained simultaneously before the design of the Drainage system is completed. Emptying time, should be as short as possible, usually about ten minutes.

Water drained from the condenser can be discharged without treatment provided that the water is kept separate and does not pass into a common drain linked to contaminated water; catchgrids are installed to remove debris. A basement sump forms a useful method of collecting the rapid discharge of water. In most cases the use of a sump means that pumping is necessary to return the water to CW channel/ Ash slurry sump/ Ash water sump. However, when the condenser is located at operating floor level gravity flow may be possible.

4.4 DRAINAGE FORM ANCILLARY PLANT

4.4.1 GENERAL SERVICE WATER TANKS

General service water tanks are usually mounted at high level and may supply water for auxiliary cooling, fire fighting and the ash plant.

4.4.2 DEAERATOR

Where the deaerator is installed at high level, a solid floor should be provided, all openings being surrounded by a small upstand. Drainage falls should be laid to gullies and drains installed to take water spillage preferably to enclosed drains in the basement since there is a likelihood of this water being hot. The deaerator may occasionally be drained and general practice is to return the water to the demineralization plant and where this is the case no extra allowance for drainage from this floor need be made.

4.4.3 WATER TREATMENT PLANT AND POLISHING PLANT

See under ``Chemical Drainage''.

4.4.4 FLOOR TRENCHES

Where trenches are used for drainage these should start at a minimum depth of 150mm below the underside of the grating. They should have a minimum fall of 1 in 200 but this should be more where possible especially where considerable quantities of solids may be discharged into them, as in the ash sluice and pulverized fuel mill areas. They should be rectangular, covered with open grids and sufficiently large to be shovelled out. Similarly, where drain pipes which are laid at self-cleansing gradients are liable to have considerable quantities of solids discharged into them, catchpits should be provided at fairly close intervals so that drain rodding may be facilitated.

It is impracticable to provided drainage to accommodate the total quantity of water at the rate of discharge if the principal fire protection drenching systems (e.g. turbo-generator) were to operate. Drainage trenches will deal with a proportion of the discharge and help prevent general flooding of areas unaffected by the fire.

5.0 DRAINAGE OF MISCELLANEOUS EXTERNAL AREAS

5.1 SURFACING AND DRAINAGE OF FUEL OIL, STORAGE AREAS

Oil storage tanks are located in an area surrounded by an earth bund or a concrete wall. Drain pipes are not laid through the bund wall to drain water from the enclosed area because valves on the pipes may be inadvertently left open and allow leaked oil to pass.

Any fuel oil delivered to a power station contains a small percentage of water which separates out from the oil and has to be drained from the inside of the tank. Water accumulates most rapidly in oil tanks filled from coastal tankers. Accumulation is very slow in tanks supplied direct by pipeline from a refinery.

5.1.1 LSHS FUEL OIL BUNDED AREAS

The Bunded area is surfaced with concrete because the oil is too viscous to soak into the ground and cause water pollution. This is only to facilitate ease of handling spilled oil.

Rainwater and water drained from inside the tanks either soaks into the ground or evaporate.

5.1.2 FUEL OIL BUNDED AREA IN COAL FIRED STATIONS (OIL TYPES HSD, LDO, & HFO)

The Bunded area is usually surfaced with concrete, because leakage of the lighter oils could soak into the ground and cause pollution and is drained to a sump within the area. The size of the sump and the slope of the floor should be sufficient for a portable pump to be used to clear the area. The drainage from each bunded area is pumped to a suitably sited oil-water separator.

5.1.3 BUNDED AREAS FOR GAS TURBINE FUEL OIL TANKS (OIL TYPE NAPHTHA)

The general provisions required for draining the Bunded area are the same as those for light oils on coal fired stations.

5.1.4 OIL PUMP HOUSE DRAINAGE

Oil transfer pumps are situated near the fuel oil tank of each type of fuel oil. Leaks may develop from the pumps and pipeline, when the station is operating. Drainage sumps for storage and subsequent pumping out must be provided unless water with leaked oil can go via oil-water separator which will retain the oil and pass the water to a convenient surface water drain. Channels collecting leaked oil should preferably be constructed in glazed vitrified clayware (since this will not absorb oil) and be covered with gratings. Pipes should not be run in drainage channels.

5.2 PRECIPITATOR AND DUST BUNKER AREAS

Normally only rainfall or water from hosing-down flows to drains from areas containing dust bunkers, ash plant, precipitators or coal handling plant and silting is likely to occur. Enclosed drainage systems should be avoided.

Under the precipitators open grid covered channels may be laid and standpipes installed at the high end to supply water for jetting down the channels. Fall, in the channels, should be as generous as possible, up to 1 in 70. Width of channels to be not less than 225 mm clear for cleaning out by shovel, away from the areas containing dust bunkers, etc. the drainage varies according to whether or not ash lagoons are within the station precincts. Where ash lagoons exist drainage is taken directly to them or to sumps which contain agitators and from here it is pumped to ash lagoons.

Where ash lagoons do not exist, the drainage is taken across catch pit from which the silt is removed.

Where much silting is likely, a catch pit large enough to be emptied by a mechanical grab should be used. Silt should, preferably, be removed from the drain near the source of contamination.

Open grid covered channels should be laid as far as possible in all other silt prone areas. Where the use of enclosed drains cannot be avoided, catch pits should be provided at the ends of each enclosed section at, say, 15m centres, so drains may be cleaned easily.

5.2.1 ASH POLLUTED DRAINAGE

A small proportion of ash will float on the surface of still water. If waterborne ash is discharged to a lagoon a crust may form giving the surface a deceptively solid appearance which could be dangerous to personnel. Scum boards should be used, in conjunction with a final settling channel.

5.3 GARAGE AREAS WHERE VEHICLES AND PLANT ARE MAINTAINED

Drainage from inside garages should pass through a ventilated petrol interceptor. Normally the drainage then flows into surface water sewers. If it is impractical to drain vehicle inspection pits, sumps should be provided and located near to gullies at floor level to facilitate baling or pumping out.

Where mobile coal-handling equipment is hosed down catch pits should be provided to prevent washed-off solids from forming blockages in the surface water drains.

5.4 OIL POLLUTED DRAINAGE

If the effluent is highly polluted by oil, and the amount of drainage is small but remote from the drainage system, it should be collected in sumps from which it is removed by road tanker.

In general, oil separators are not required to deal with heavy oil which should be allowed to separate in sumps. Drainage from plant areas which could be contaminated by light oil should pass through an oil separator of the American Petroleum Institute (A.P.I) or tilted plate type before discharge. Oil interceptors only work effectively under smooth flow conditions so it may be advisable to provide interceptor chambers or a holding reservoir to deal with turbulent or intermittent discharges before the water passed through the separator.

Oil seepage to drains at a sub-station may be adequately dealt with by separators similar to the standard petrol interceptor.

Solid particles should be dealt with as described for precipitator and dust bunker areas before the water enters the separator. This will prevent a build up occurring in the separator, which tends to act as a catch pit.

6.0 CHEMICAL DRAINAGE

6.1 WATER TREATMENT PLANT

The effluent discharged from the polishing plant is generally within the range 10 % acid to 10% alkali. It should be drained to the effluent holding and neutralizing sump, the drains having a suitable chemical resistance.

Drainage from the acid and alkali bulk storage can be dealt with by one of the following methods:

a) By drains to the effluent holding and neutralizing sump

b) By a sump located within the Bunded areas. In this case spillage is neutralized in the sump and then pumped to a foul water drain.

The effluent holding and neutralizing sump (Bulk effluent tank) and the drains leading to it should have a chemical resistant lining suitable for the worst condition, which is likely to be discharge of acid or alkali from the bulk storage tanks without neutralization.

The discharge from the panic shower is very diluted and chemical resistant drainage is not required.

A chemical resistant sink or sump is provided below the tanker coupling points to chemical bulk storage tanks and any spilt chemicals are washed away into foul water system. Special drains are not required.

6.1.1 STATION LABORATORIES

White glazed fireclay sinks in laboratories should be fitted with polyethylene receivers or catch pits which retain a reasonable quantity of waste liquid to dilute any chemical poured into the sink. Large quantities of acids and alkalis are not used in station laboratories and five litre receivers should be adequate to ensure dilution. The receivers should be connected to the foul drainage system by thermoplastic pipework.

7.0 UPKEEPING AND REPAIR/MAINTENANCE RESPONSIBILITY

Routine cleaning/ Up-keeping/ housekeeping of the internal drains (covered/ open type as per the attached sketches) trenches/ siphons/ catch pit/ gulley trap/ manholes of the different areas shall be taken up by the respective departments as detailed below:

Sl. No.	Location/Areas	Frequency of Check	Responsibility
1	Main plant Stage- I & II	Every year pre & post monsoon	Operation
2	WTP area	do	O&M-Chemistry
3	AHP area	do	MM-Ash handing dept.
4	Fly Ash Silo	do	Operation
5	CHP area	do	CHP
6	Cooling Towers	do	O&M-Civil
7	Stores	do	Stores
8	Switchyard & Switchgear	do	EMD
9	Other building & Pump house	do	Concerned house keeping group

Repair/rerouting of all drains shall be taken up by O&M-Civil dept. as per site condition/requirement basis.

8.0 EXCEPTIONAL REPORTING

Exception report findings of pre monsoon committee to be complied by respective departments.

9.0 REVIEW

This document shall be reviewed by Head of Simhadri every two years or earlier if required.

NTPC			LMI	SIMHADRI
LMI FOR POWER STATION DRAINAGE				
REVISION INDEX				
Sl. No.	**Rev. No**	**Date**	**Details**	**Approved by**
1	0	May’2019		HOP (Simhadri)
2	1	May’2021		HOP (Simhadri)
3	2	Aug’2022		HOP (Simhadri)

NTPC		LMI	SIMHADRI
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